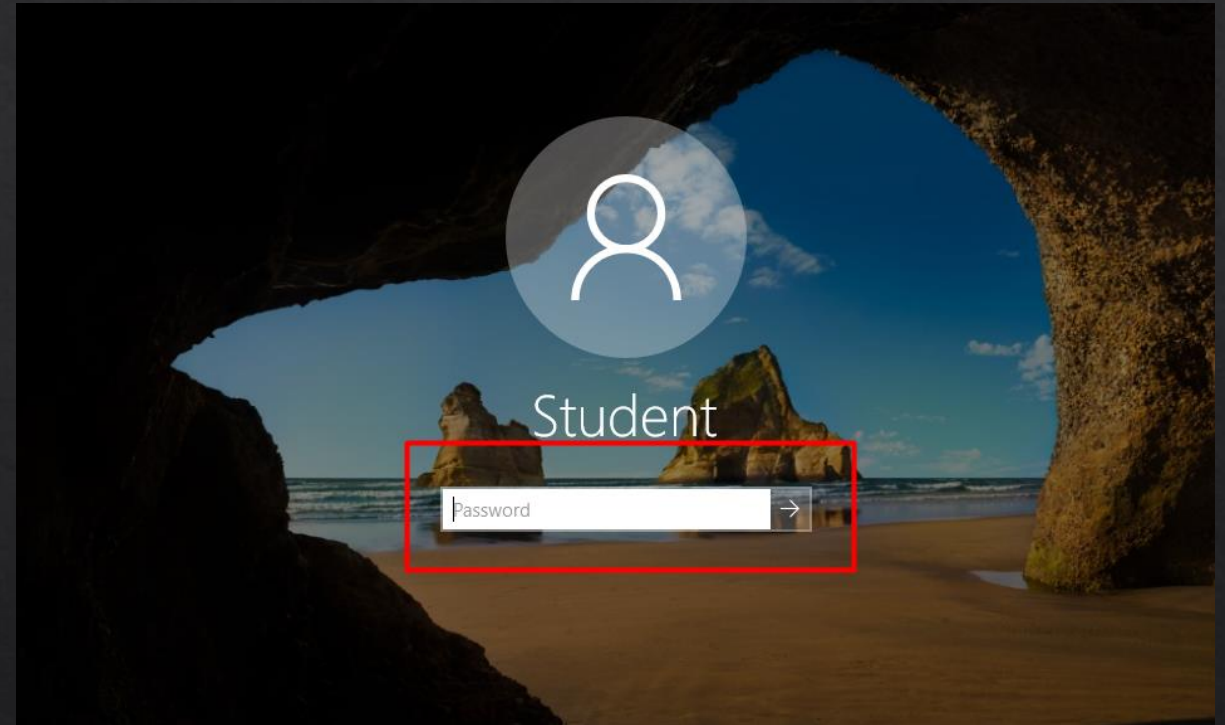
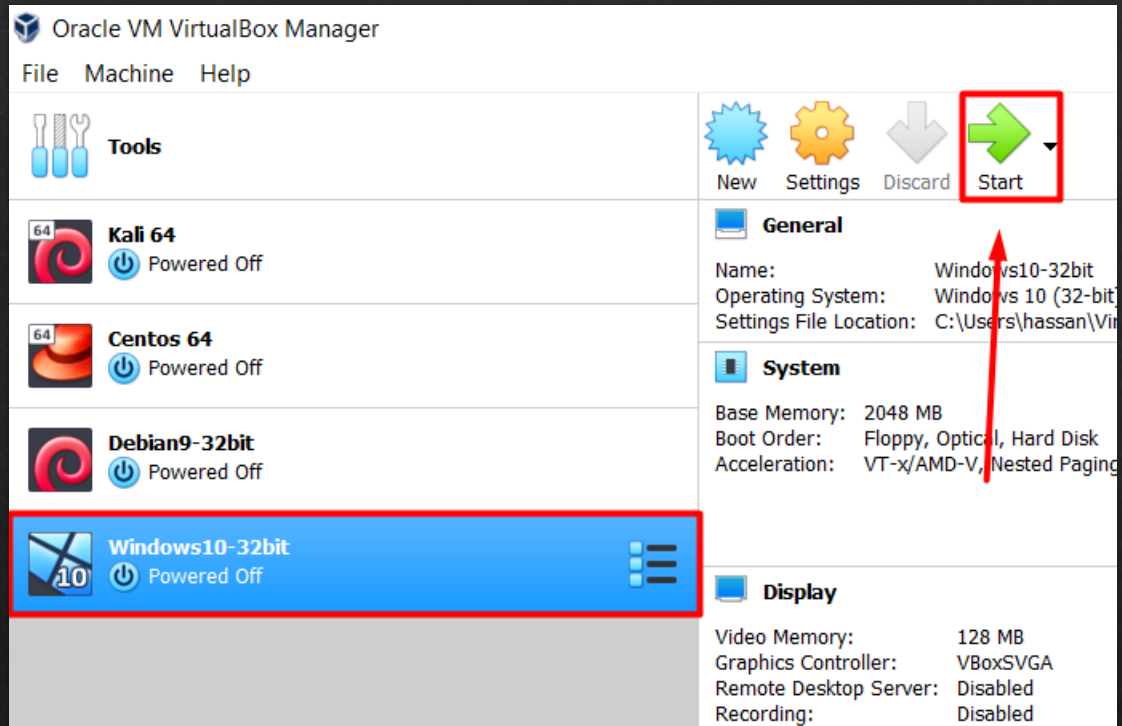


Introduction To Windows

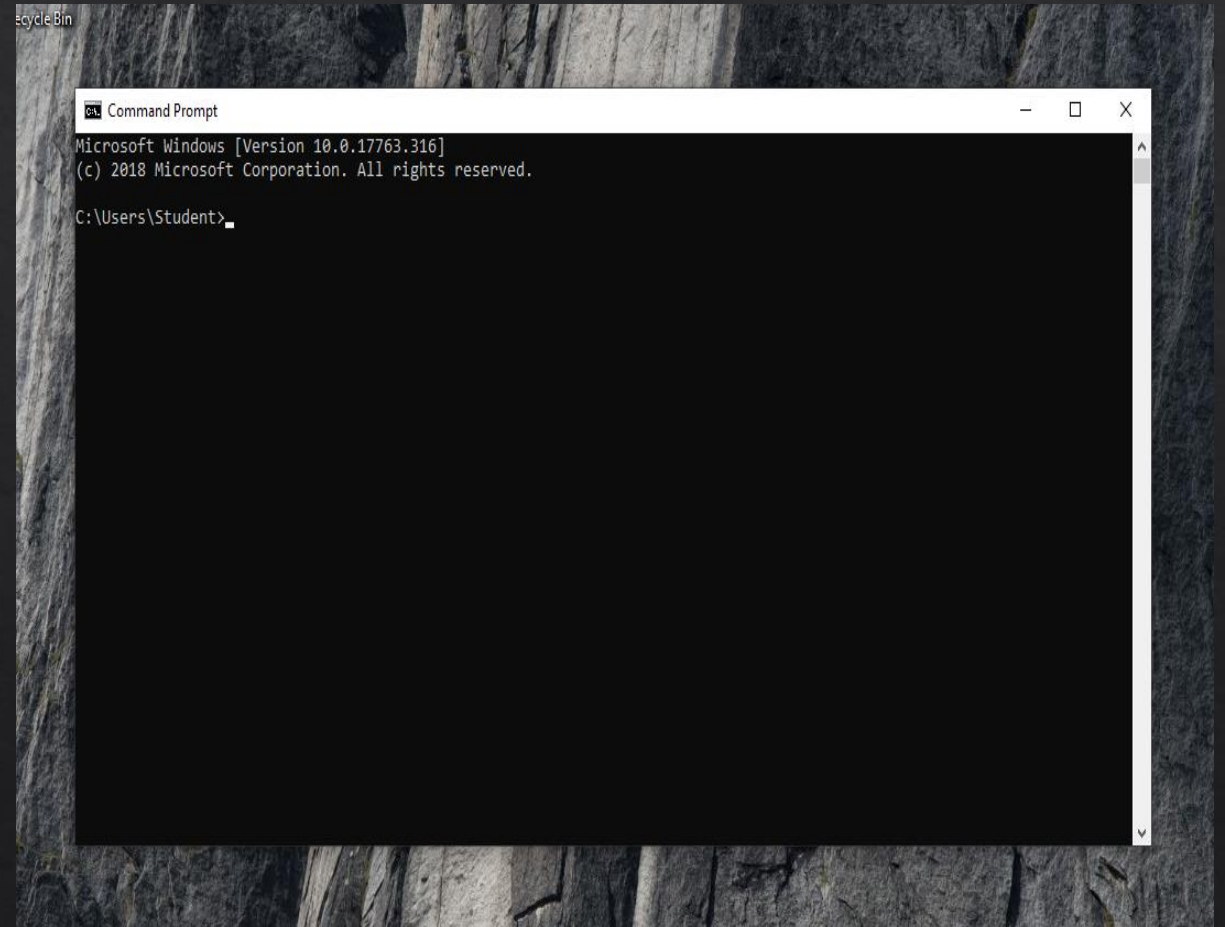
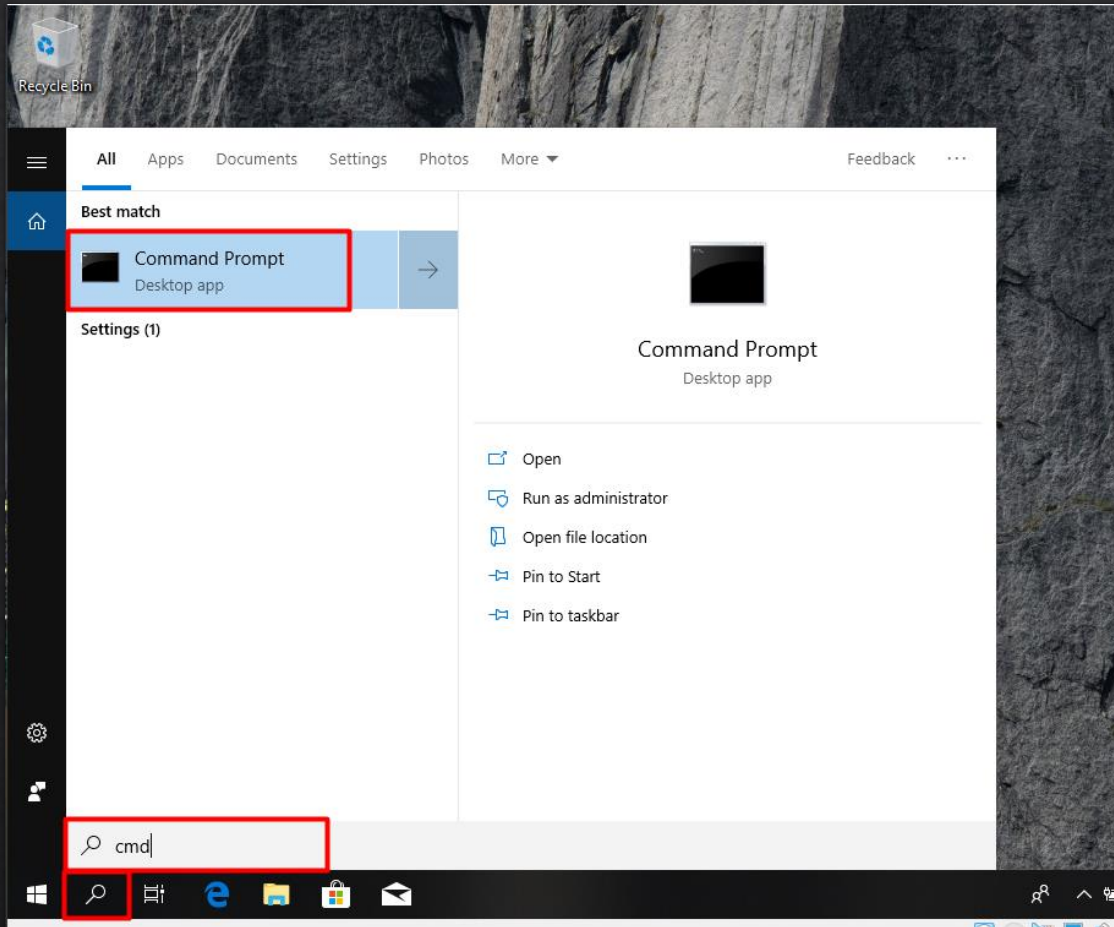
Run Windows



Username: student

Password: student

Run Windows



Core Commands

Command	Description	Example			
dir	List folders and files	dir	dir C:\Users	dir /a	dir /a C:\Users
cd	Change directory	E: C:	cd C:\Users	cd ..	cd Desktop
mkdir	Create directory	mkdir test			
rmdir	Remove directory	rmdir test		rmdir /s non-empty-test	
move	Move files or folders	move file1.txt folder1	move a.txt b.txt	move folder1 folder2	
echo	Print text	echo “Hello World”		Echo “test” > file1.txt	
type	View content of a file	type file1.txt			
del	Delete file	del file1.txt			
copy	Copy file	copy a.txt b.txt		copy a.txt folder1	

Core Commands

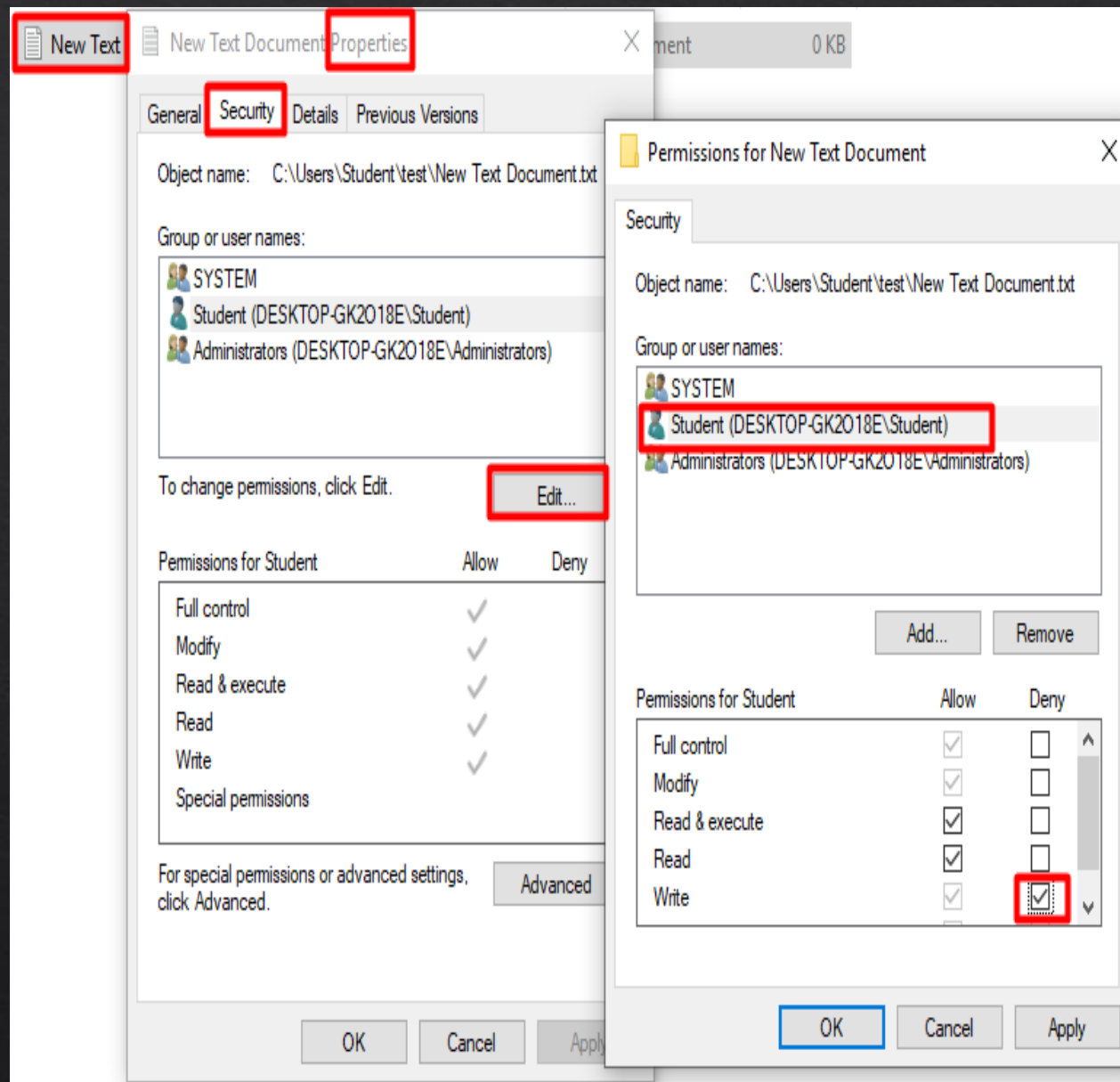
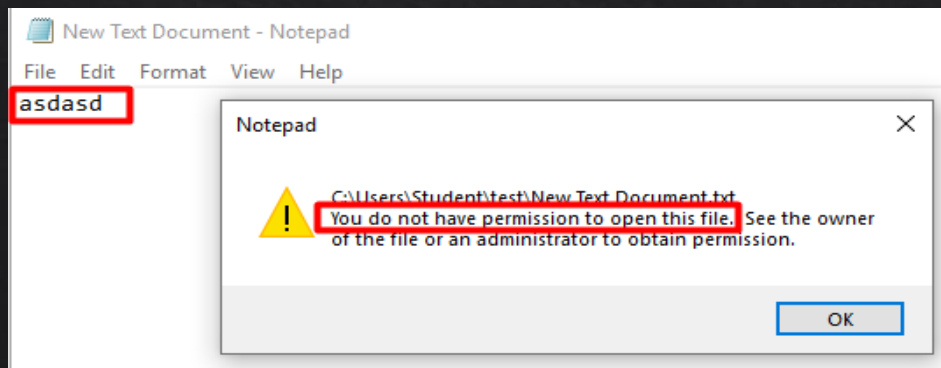
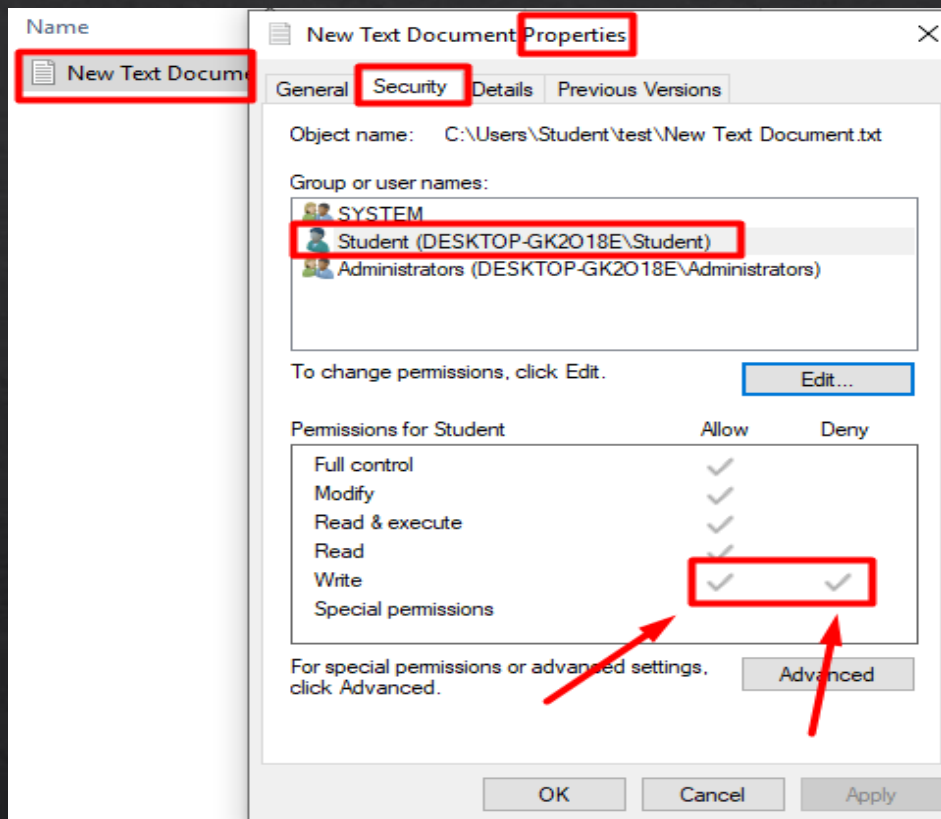
Command	Description	Example			
ren	Rename file	ren a.txt newname.txt			
attrib	Change attributes of files or folders	attrib +h a.txt	attrib -h a.txt	attrib +h folder	dir /a:h
tasklist	List running processes	tasklist			
taskkill	Kill running process	taskkill /f /pid 1547			
ipconfig	Display network information	ipconfig		ipconfig /all	
netstat	Display active connections	netstat		netstat /ano	
ping	Test network connectivity	ping google.com			
tracert	Show path to destination	tracert google.com			
mklink	Create links	mklink soft file	mklink /H hard file		mklink /D link dir

Windows File System

Folder name	Folder description
C:\	Top level point that hold all the system files.
C:\Windows	Contains operating system files.
C:\Program Files	Contains applications files.
C:\Program Files (x86)	Contains 32bit applications on 64bit operating system.
C:\Users	Contains users home directories.
C:\ProgramData	Contains configuration files of applications (Hidden).

File Permissions

- ◇ Windows has 5 main permission types:
 - ◇ Read, Write, Full Control, Read & Execute, Modify.
- ◇ Each user and group on the system has its own permissions:
 - ◇ User1: Read, User2: Read + Write, Group1: Full Control.
- ◇ Inherited permissions:
 - ◇ Child inherit parent permissions. (This is what happens by default in windows)
- ◇ Explicit permissions:
 - ◇ Permission applied specially to a file.
- ◇ **Explicit Deny > Explicit Allow > Inherited Deny > Inherited Allow.**



Users & Groups

- ◆ Create user:
 - ◆ net user testuser * /add
- ◆ Delete user:
 - ◆ net user testuser /delete
- ◆ Disable/Enable user account:
 - ◆ net user testuser /active:no
 - ◆ net user testuser /active:yes
- ◆ Display password policy:
 - ◆ net accounts

```
Administrator: Command Prompt
C:\Windows\system32>net user testuser * /add
Type a password for the user:
Retype the password to confirm:
The command completed successfully.

C:\Windows\system32>net user testuser /active:no
The command completed successfully.

C:\Windows\system32>net user testuser /active:yes
The command completed successfully.

C:\Windows\system32>net user testuser /delete
The command completed successfully.

C:\Windows\system32>net accounts
Force user logoff how long after time expires?: Never
Minimum password age (days): 0
Maximum password age (days): 42
Minimum password length: 0
Length of password history maintained: None
Lockout threshold: Never
Lockout duration (minutes): 30
Lockout observation window (minutes): 30
Computer role: WORKSTATION
The command completed successfully.
```

Users & Groups

- ◆ Most used groups:
 - ◆ Administrators: Full control over system.
 - ◆ Network Operators : Allow to modify network settings (IP, DNS).
 - ◆ Users: Allow access to needed functionality by most users.
- ◆ Display groups:
 - ◆ net localgroup
- ◆ List group members:
 - ◆ net localgroup administrators

```
C:\Windows\system32>net localgroup

Aliases for \\DESKTOP-GK2018E

-----
*Access Control Assistance Operators
*Administrators
*Backup Operators
*Cryptographic Operators
*Device Owners
*Distributed COM Users
*Event Log Readers
*Guests
*Hyper-V Administrators
*ITS_USERS
*Network Configuration Operators
*Performance Log Users
*Performance Monitor Users
*Power Users
*Remote Desktop Users
*Remote Management Users
*Replicator
*System Managed Accounts Group
*Users
The command completed successfully.

C:\Windows\system32>net localgroup administrators

Alias name     administrators
Comment       Administrators have complete and unre
Members

-----
Administrator
Student
The command completed successfully.
```

Users & Groups

- ◆ Create group:
 - ◆ net localgroup testgroup /add
- ◆ Add user to group:
 - ◆ net localgroup testgroup testuser /add
- ◆ Delete user from group:
 - ◆ net localgroup testgroup testuser /del
- ◆ Delete group:
 - ◆ net localgroup testgroup /del

```
C:\Windows\system32>net localgroup testgroup /add
The command completed successfully.
```

```
C:\Windows\system32>net localgroup testgroup testuser /add
The command completed successfully.
```

```
C:\Windows\system32>net localgroup testgroup
Alias name     testgroup
Comment
```

```
Members
```

```
-----
testuser
The command completed successfully.
```

```
C:\Windows\system32>net localgroup testgroup testuser /del
The command completed successfully.
```

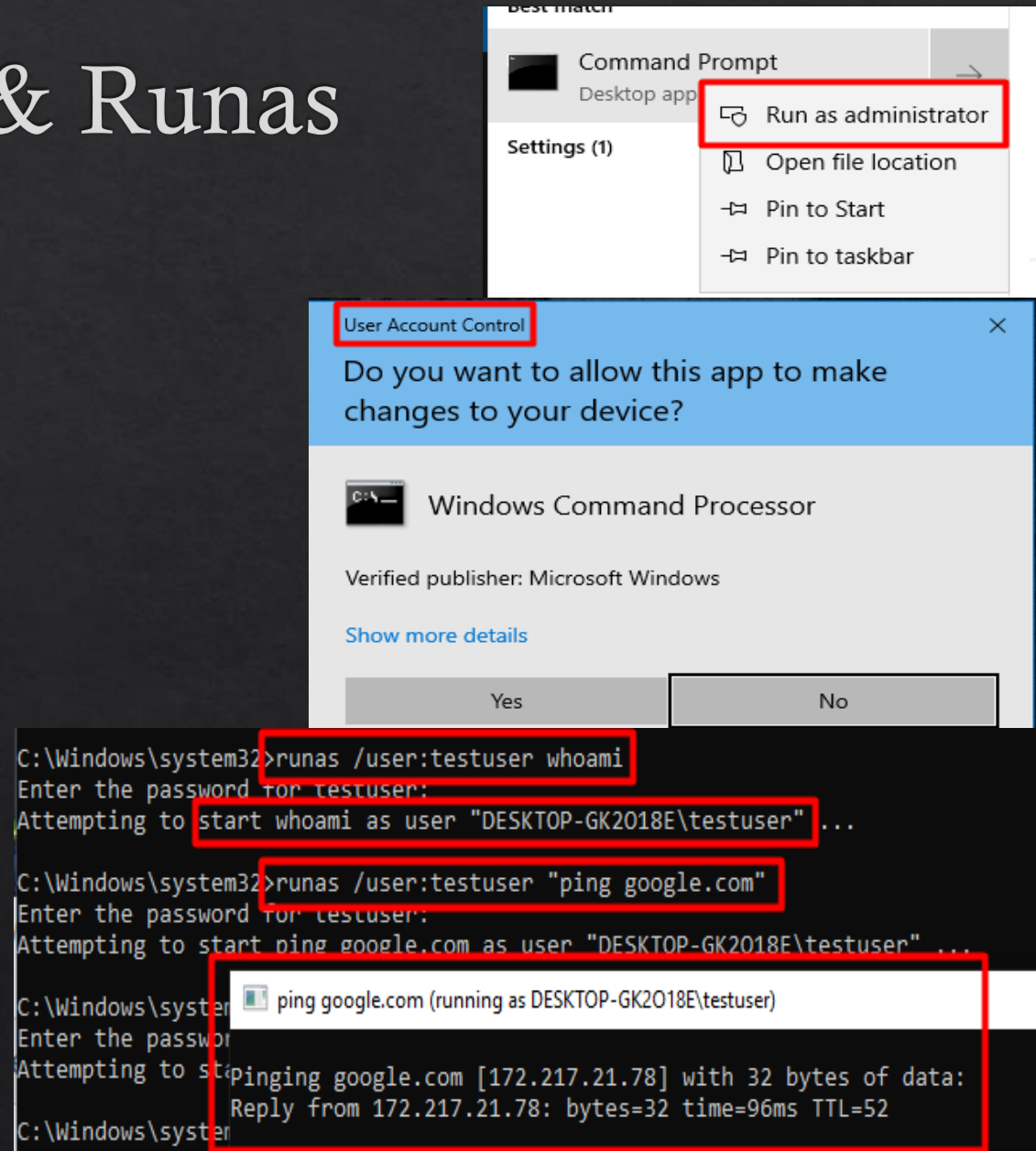
```
C:\Windows\system32>net localgroup testgroup /del
The command completed successfully.
```

```
C:\Windows\system32>net localgroup testgroup /del
System error 1376 has occurred.
```

```
The specified local group does not exist.
```

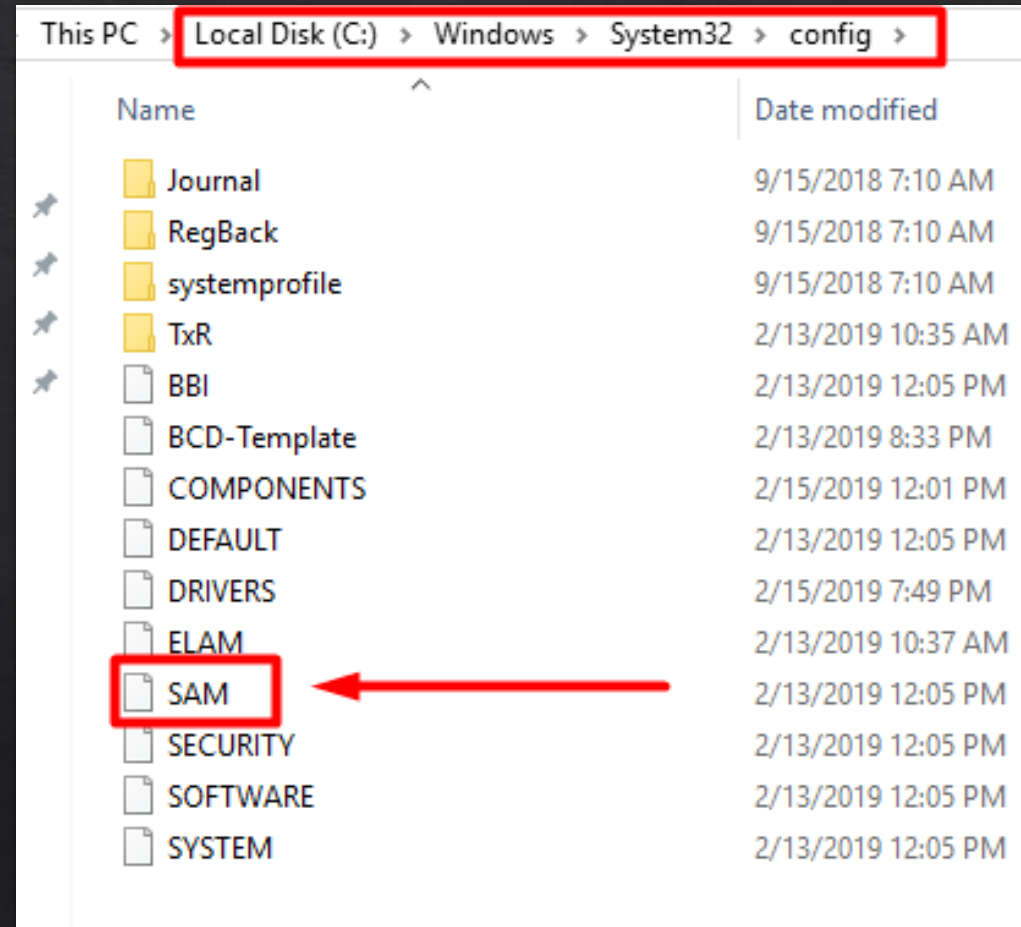
UAC & Runas

- ❖ UAC gives you administrator access for one command, you trigger it by clicking right click on any file then choose run as administrator.
- ❖ runas lets you run command as another user like in linux (su testuser -c whoami):
 - ❖ runas /user:testuser whoami
 - ❖ runas /user:testuser "ping google.com"



Credentials

- ◆ Credentials (usernames & passwords) are stored in the SAM file.
- ◆ SAM file location:
 - ◆ C:\Windows\System32\config.
- ◆ Mostly stores the users' passwords in the NTLM hash.

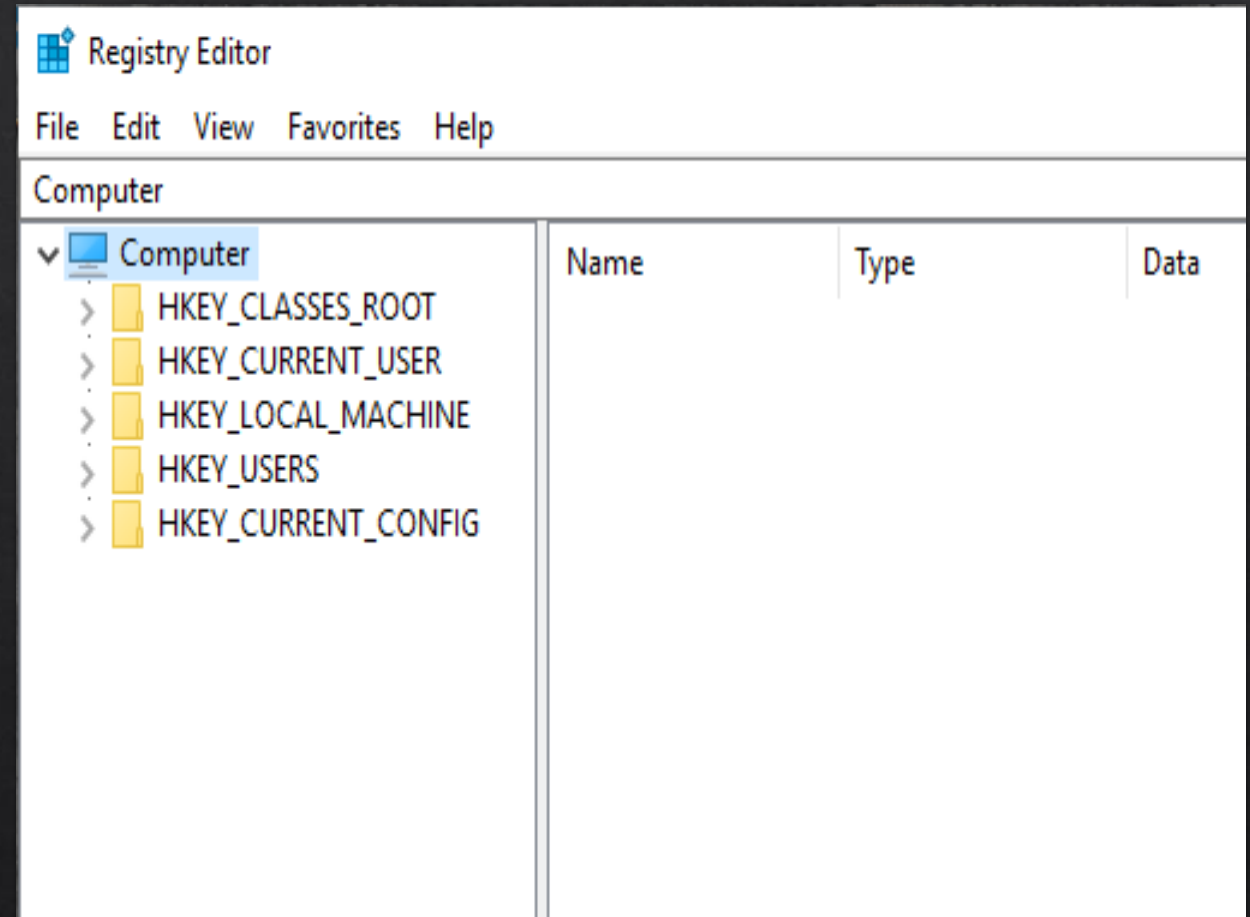


Security Policy

- ◇ Audit policy:
 - ◇ Control logging through event viewer.
- ◇ User rights:
 - ◇ Control permissions on the OS (change time, backup).
- ◇ Security options:
 - ◇ password length.
 - ◇ password complexity.
 - ◇ password expiration period.
 - ◇ rename administrator account.

Registry

- ◆ Registry stores configuration settings, and there are 5 types of registry hives:
 - ◆ HKEY_Local_Machine (HKLM)
 - ◆ local machine settings.
 - ◆ HKEY_USERS (HKU)
 - ◆ Settings for every user.
 - ◆ HKEY_Current_User (HKCU)
 - ◆ settings for the current user.
 - ◆ HKEY_Current_Config (HKCC)
 - ◆ settings for hardware.
 - ◆ HKEY_Classes_Root (HKCR)
 - ◆ settings for applications.



Registry

- ◆ Print help:
 - ◆ reg /?
- ◆ Print help on specific function:
 - ◆ reg query /?
- ◆ Query specific registry hives:
 - ◆ reg query hkcu
 - ◆ reg query hklm
 - ◆ reg query hkcu\software
- ◆ Print all startup apps recursively:
 - ◆ reg query hklm\software\microsoft\windows\currentversion\run /s

```
Administrator: Command Prompt

C:\Windows\system32>reg query hkcu

HKEY_CURRENT_USER\AppDataEvents
HKEY_CURRENT_USER\Console
HKEY_CURRENT_USER\Control Panel
HKEY_CURRENT_USER\Environment
HKEY_CURRENT_USER\EUDC
HKEY_CURRENT_USER\Keyboard Layout
HKEY_CURRENT_USER\Network
HKEY_CURRENT_USER\Printers
HKEY_CURRENT_USER\SOFTWARE
HKEY_CURRENT_USER\SYSTEM
HKEY_CURRENT_USER\Volatile Environment

C:\Windows\system32>reg query hkcu\software

HKEY_CURRENT_USER\software\AppDataLow
HKEY_CURRENT_USER\software\Google
HKEY_CURRENT_USER\software\Microsoft
HKEY_CURRENT_USER\software\Policies
HKEY_CURRENT_USER\software\RegisteredApplications
HKEY_CURRENT_USER\software\Classes

C:\Windows\system32>reg query hklm\software\microsoft\windows\currentversion\run /s

HKEY_LOCAL_MACHINE\software\microsoft\windows\currentversion\run
SecurityHealth REG_EXPAND_SZ %windir%\system32\SecurityHealthSystray.exe
VBoxTray REG_SZ C:\Windows\system32\VBoxTray.exe
```

Registry

- ◆ Add startup key called EvilTest which started the Calculator app at boot time:
 - ◆ reg add
hklm\software\microsoft\windows\currentversion\run /v "EvilTest" /d "calc.exe"
- ◆ Delete EvilTest key:
 - ◆ reg delete
hklm\software\microsoft\windows\currentversion\run /v "EvilTest"

```
Administrator: Command Prompt

C:\Windows\system32>reg add hklm\software\microsoft\windows\currentversion\run /v "EvilTest" /d "calc.exe"
The operation completed successfully.

C:\Windows\system32>reg query hklm\software\microsoft\windows\currentversion\run /s

HKEY_LOCAL_MACHINE\software\microsoft\windows\currentversion\run
    SecurityHealth    REG_EXPAND_SZ    %windir%\system32\SecurityHealthSystray.exe
    VBoxTray         REG_SZ           C:\Windows\system32\VBoxTray.exe
    EvilTest          REG_SZ           calc.exe

C:\Windows\system32>reg delete hklm\software\microsoft\windows\currentversion\run /v "EvilTest"
Delete the registry value EvilTest (Yes/No)? Yes
The operation completed successfully.

C:\Windows\system32>reg query hklm\software\microsoft\windows\currentversion\run /s

HKEY_LOCAL_MACHINE\software\microsoft\windows\currentversion\run
    SecurityHealth    REG_EXPAND_SZ    %windir%\system32\SecurityHealthSystray.exe
    VBoxTray         REG_SZ           C:\Windows\system32\VBoxTray.exe

C:\Windows\system32>
```

Windows Sharing (SMB)

- ◆ Print help:
 - ◆ net /?
- ◆ List all devices that is enable sharing on the network:
 - ◆ net view
- ◆ list all shared resources on specific system:
 - ◆ net view \\computer-name
- ◆ mapping resource to drive letter:
 - ◆ net use z: \\computer-name\resource
- ◆ show mapping resources:
 - ◆ net use
- ◆ delete mapping resource:
 - ◆ net use z: /delete

```
C:\Windows\system32>net view \\127.0.0.1
Shared resources at \\127.0.0.1
```

```
Share name  Type  Used as  Comment
-----
```

```
C      Disk
The command completed successfully.
```

```
C:\Windows\system32>net use z: \\127.0.0.1\c
The command completed successfully.
```

```
C:\Windows\system32>net use
New connections will be remembered.
```

Status	Local	Remote	Network
OK	Z:	\\127.0.0.1\c	Microsoft Windows Network

```
The command completed successfully.
```

```
C:\Windows\system32>net use z: /delete
z: was deleted successfully.
```

```
C:\Windows\system32>net use
New connections will be remembered.
```

```
There are no entries in the list.
```

Services

- ◆ Display active services:
 - ◆ sc query
- ◆ Display info about spooler service:
 - ◆ sc query spooler
- ◆ Display inactive services:
 - ◆ sc query state=inactive
- ◆ Display all services:
 - ◆ sc query state=all
- ◆ Start/Stop the spooler service:
 - ◆ sc start spooler
 - ◆ sc stop spooler
- ◆ Enable/Disable spooler service at boot time:
 - ◆ sc config spooler start=disabled
 - ◆ sc config spooler start=auto

```
C:\Windows\system32>sc query spooler

SERVICE_NAME: spooler
        TYPE               : 110    WIN32_OWN_PROCESS (interactive)
        STATE                : 4      RUNNING
                                (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0      (0x0)
        SERVICE_EXIT_CODE    : 0      (0x0)
        CHECKPOINT            : 0x0
        WAIT_HINT             : 0x0

C:\Windows\system32>sc start spooler
[SC] StartService FAILED 1056:

An instance of the service is already running.

C:\Windows\system32>sc stop spooler

SERVICE_NAME: spooler
        TYPE               : 110    WIN32_OWN_PROCESS (interactive)
        STATE                : 3      STOP_PENDING
                                (NOT_STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0      (0x0)
        SERVICE_EXIT_CODE    : 0      (0x0)
        CHECKPOINT            : 0x3
        WAIT_HINT             : 0x4e20

C:\Windows\system32>sc config spooler start=disabled
[SC] ChangeServiceConfig SUCCESS

C:\Windows\system32>sc config spooler start=auto
[SC] ChangeServiceConfig SUCCESS

C:\Windows\system32>
```

Processes

- ◆ List all tasks:
 - ◆ tasklist
- ◆ Find task by it's name or id:
 - ◆ tasklist /fi "imagename eq calc*"
- ◆ Force kill process:
 - ◆ taskkill /f /pid 4054

```
Administrator: Command Prompt

C:\Windows\system32>calc.exe

C:\Windows\system32>tasklist /fi "imagename eq calc*"

Image Name                PID Session Name        Session#    Mem Usage
-----
Calculator.exe            780 Console                1         40,124 K

C:\Windows\system32>taskkill /f /pid 780
SUCCESS: The process with PID 780 has been terminated.

C:\Windows\system32>
```

Processes

- ◆ Create calculator process:
 - ◆ wmic process call create calc.exe
- ◆ Search for calculator process:
 - ◆ wmic process where (name = "calculator.exe") list brief
 - ◆ wmic process where (name like "calc%") list brief
- ◆ Delete calculator process:
 - ◆ wmic process where (name = "calculator.exe") delete

```
C:\Windows\system32>wmic process call create calc.exe
Executing (Win32_Process)->Create()
Method execution successful.
Out Parameters:
instance of __PARAMETERS
{
    ProcessId = 780;
    ReturnValue = 0;
};

C:\Windows\system32>wmic process where (name="calculator.exe") list brief
HandleCount  Name                Priority  ProcessId  ThreadCount  WorkingSetSize
570          Calculator.exe      8        5024       24           41222144

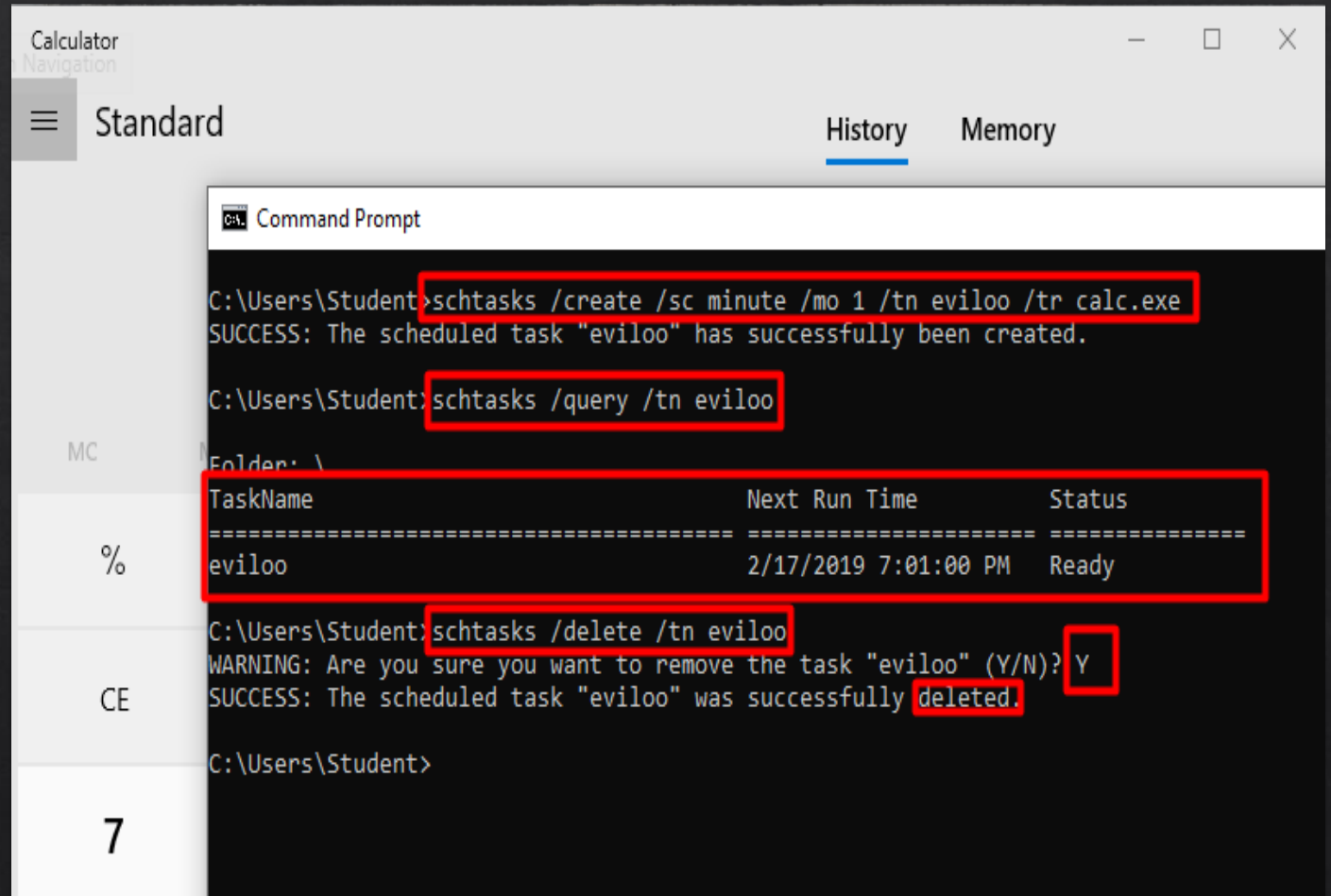
C:\Windows\system32>wmic process where (name like "calc%") list brief
HandleCount  Name                Priority  ProcessId  ThreadCount  WorkingSetSize
570          Calculator.exe      8        5024       24           41222144

C:\Windows\system32>wmic process where (name="calculator.exe") delete
Deleting instance \\DESKTOP-GK2018E\ROOT\CIMV2:Win32_Process.Handle="5024"
Instance deletion successful.

C:\Windows\system32>
```

Task Scheduling

- ◆ Display all scheduled tasks:
 - ◆ schtasks
- ◆ Create new task to run every minute:
 - ◆ schtasks /create /sc minute /mo 1 /tn eviloo /tr calc.exe
- ◆ Display information about specific scheduled task:
 - ◆ schtasks /query /tn eviloo
- ◆ Delete scheduled task:
 - ◆ schtasks /delete /tn eviloo



The screenshot shows a Windows Calculator window in the background. Overlaid on top is a Command Prompt window. The Command Prompt shows the following commands and their outputs:

```
C:\Users\Student>schtasks /create /sc minute /mo 1 /tn eviloo /tr calc.exe
SUCCESS: The scheduled task "eviloo" has successfully been created.

C:\Users\Student>schtasks /query /tn eviloo
```

TaskName	Next Run Time	Status
eviloo	2/17/2019 7:01:00 PM	Ready

```
C:\Users\Student>schtasks /delete /tn eviloo
WARNING: Are you sure you want to remove the task "eviloo" (Y/N)? Y
SUCCESS: The scheduled task "eviloo" was successfully deleted.

C:\Users\Student>
```

The calculator window in the background has tabs for 'Standard', 'History', and 'Memory'. The 'Standard' tab is active, showing a navigation pane on the left with 'MC', '%', 'CE', and '7'.